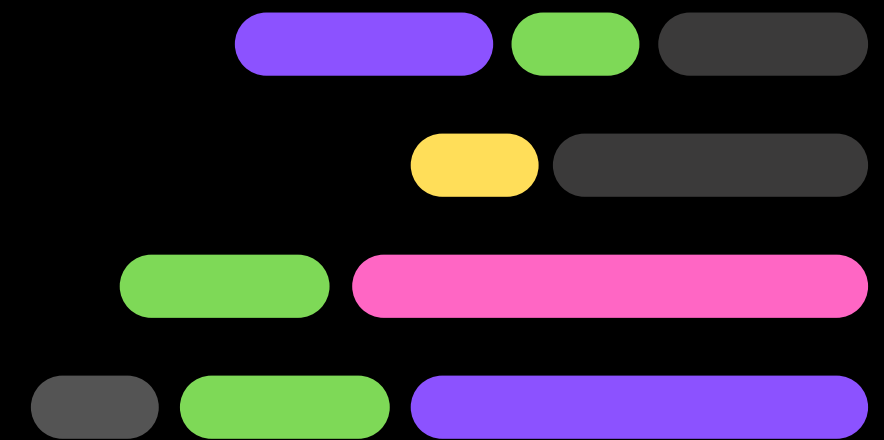
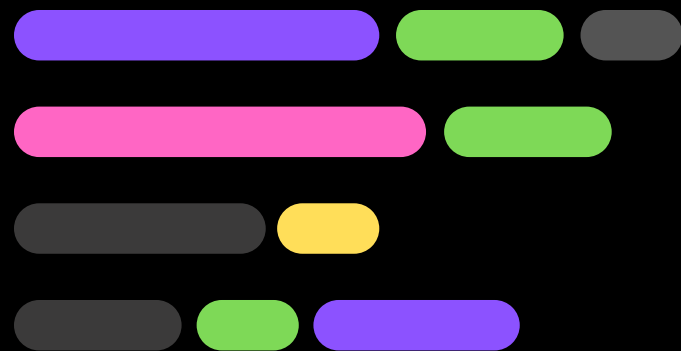




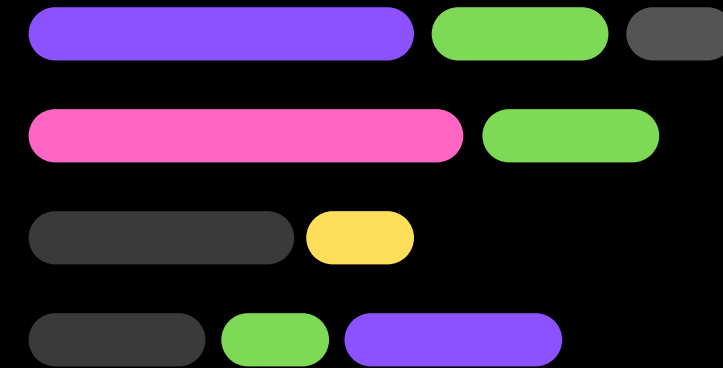
{Python Programming}

CodeStarters

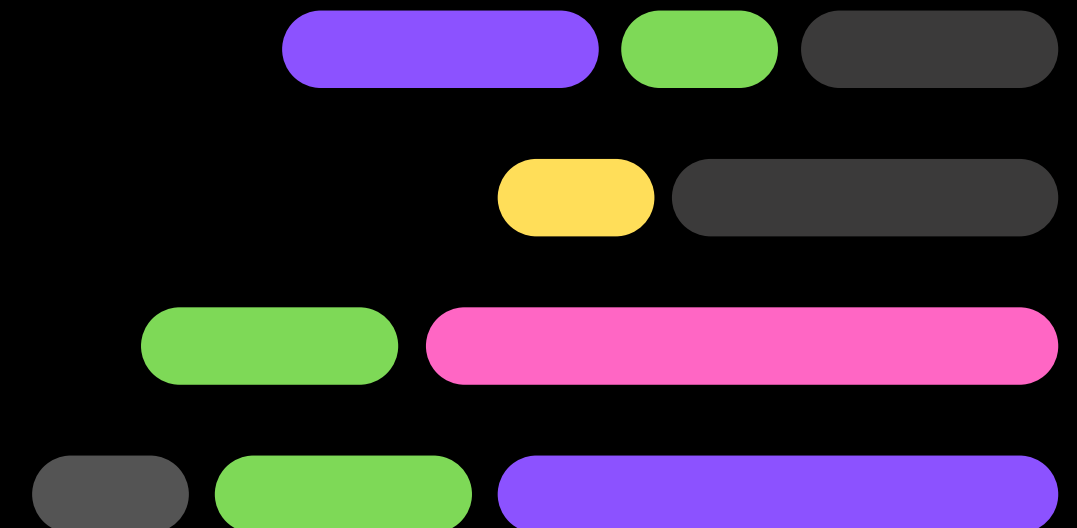
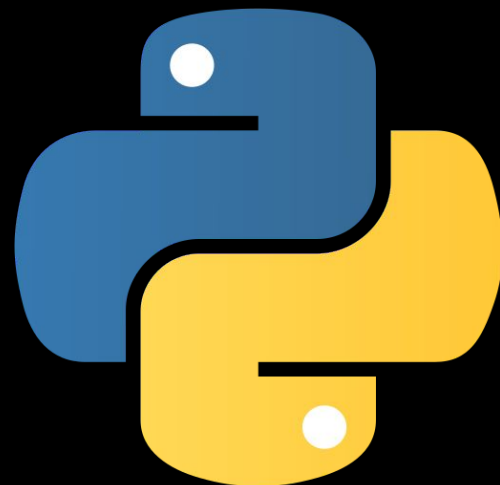




Review



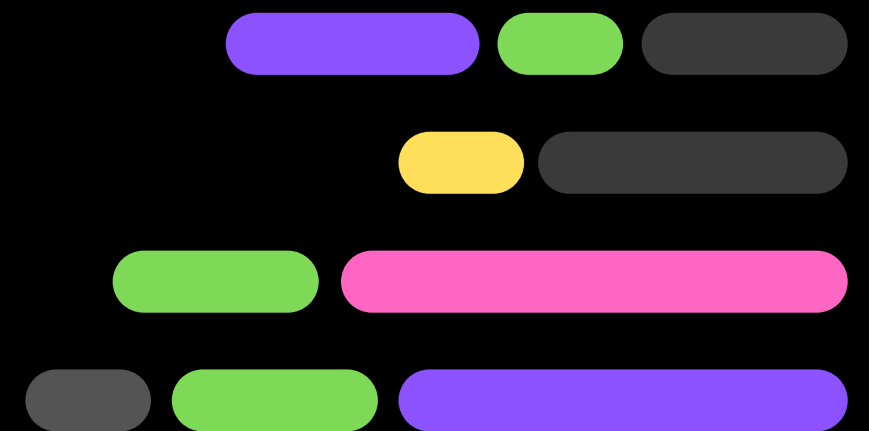
If/Else Statements





Review

Booleans - True / False





Lists

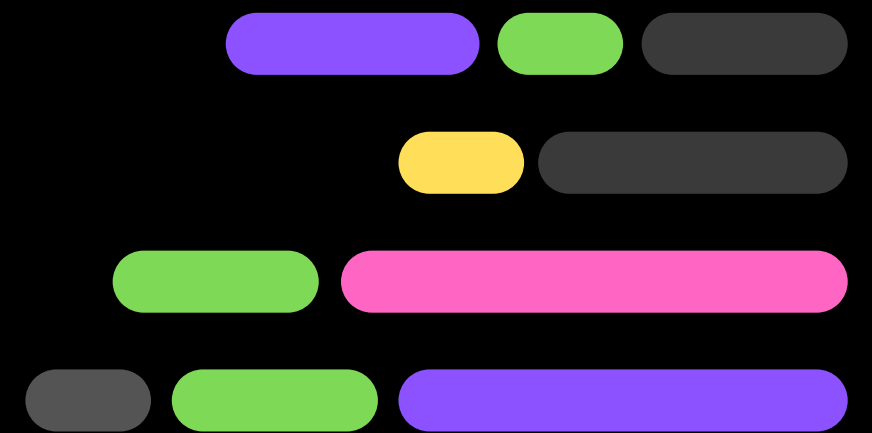
What are lists?



Programming Concepts

Lists

Appending and Indexing





List

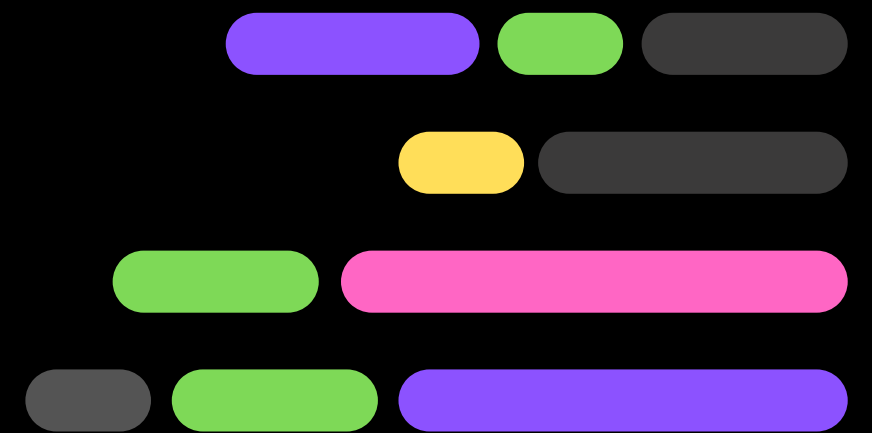
Exercise



Programming Concepts

Functions

Code that you want
to reuse

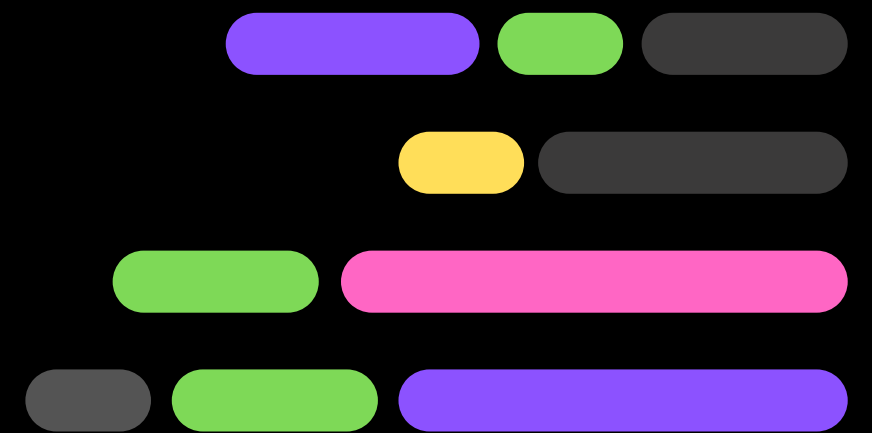




Programming Concepts

FUNCTIONS

- Return values
- Parameters





Functions

Exercise

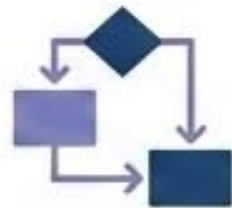
Programming Concepts



Loops

```
for i in range(5):
```

Repeat a block of code multiple times.



Conditional Statements

```
if score > 90:
```

Execute different code paths based on logic.



Variables

```
user_name = "Alex"
```

Store data values.



Strings

```
"Hello, World!"
```

Handle text data.



Functions

```
def greet():
```

Reuse code blocks by defining them.



Function Challenges

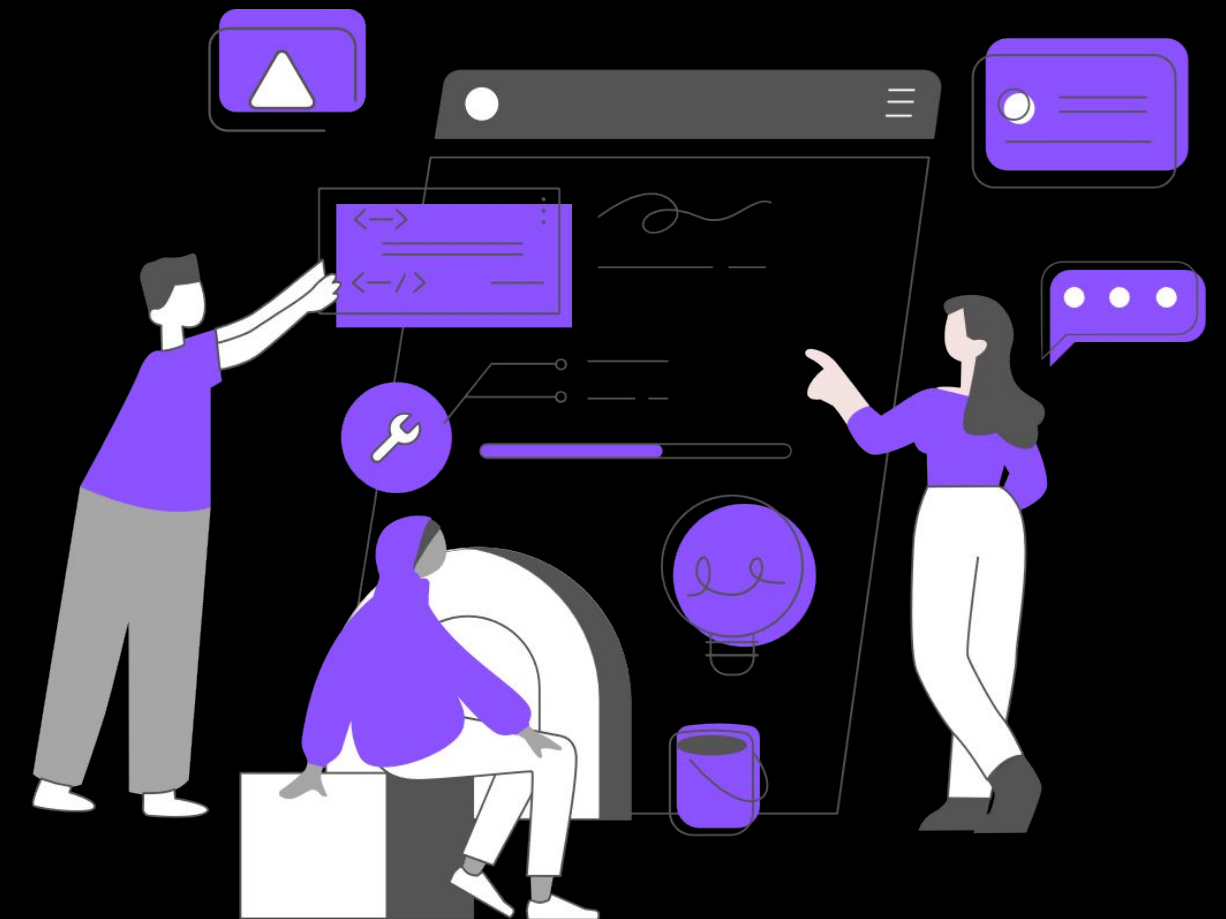
- 1) **Secret Word** - A function checks if user guesses the secret word.
- 2) **Find Biggest Number** - Function returns the larger number.
- 3) **Multiply Machine** - Function multiplies two numbers.
- 4) **Tip Calculator** - Function returns the total including a tip.
- 5) **Area of Triangle** - Function takes base and height and returns the area of the triangle.

```
public class Calculator {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt user for input  
        System.out.print("Enter the first number: ");  
        double num1 = scanner.nextDouble();  
  
        System.out.print("Enter the second number: ");  
        double num2 = scanner.nextDouble();  
  
        System.out.print("Enter the operation (+, -, *, /): ");  
        char operation = scanner.next().charAt(0);  
  
        double result = 0;
```

Final Project!

You will have a final project and work time will be giving for the last two days! The project must have a **working demo** and you should **present** your project at the end of the week. The project should be **text based on the terminal**. Ideas include **games, quizzes**, etc.

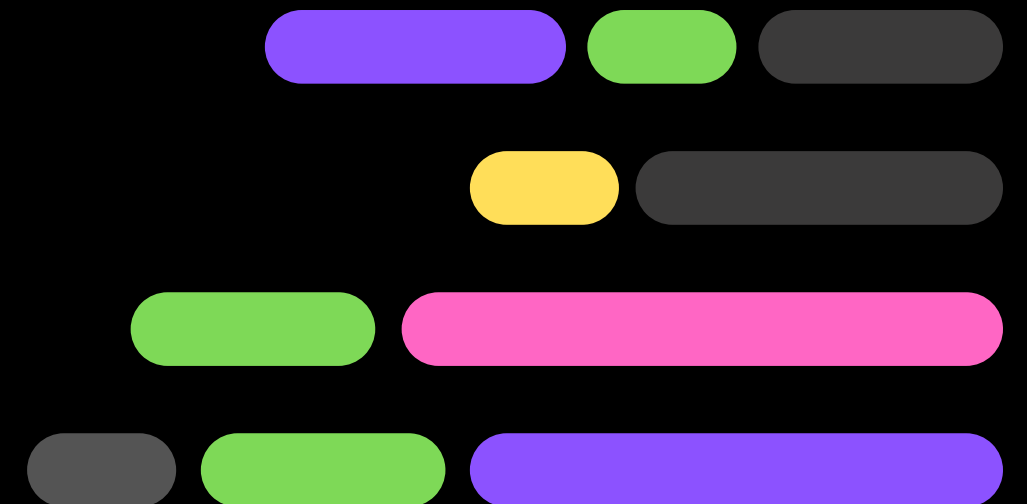
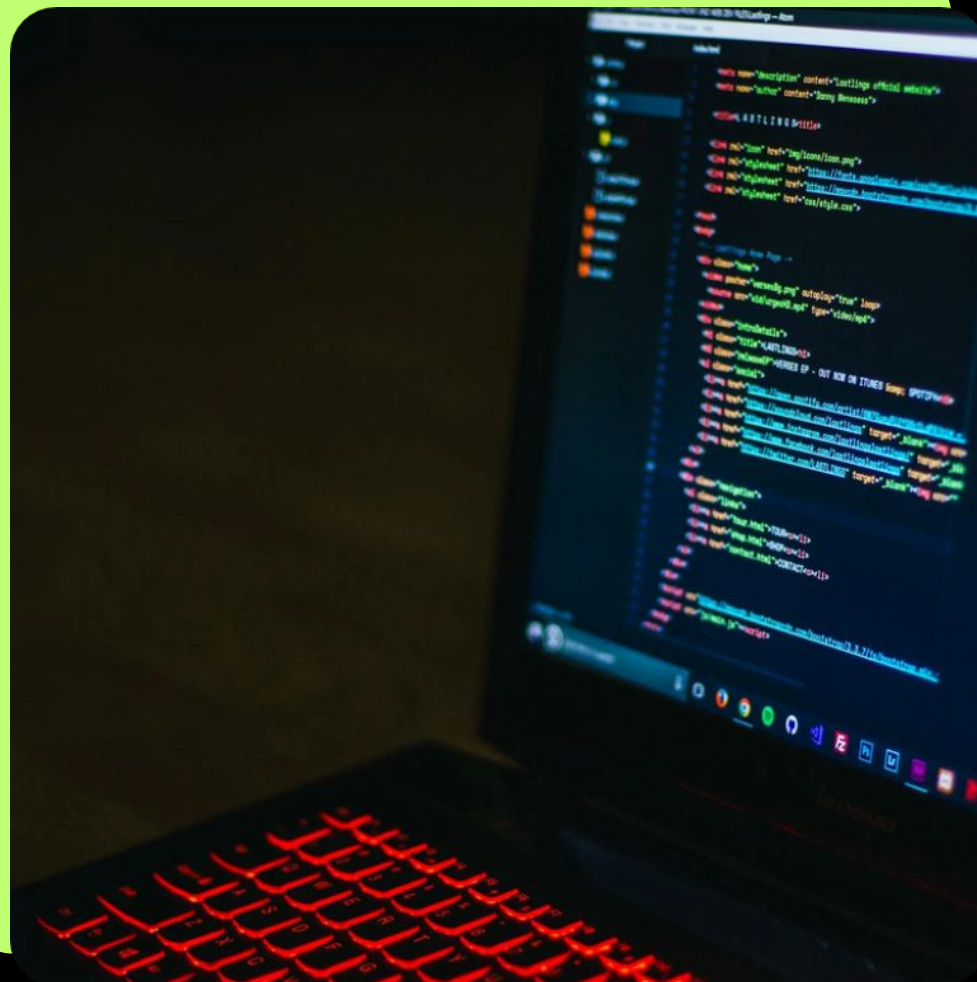
For the **requirements**: the code must be **shared**, you must **take the user's input** for the code to function, and there must be **tracking of data** (variables or lists).





Brainstorm

This is time to brainstorm ideas!





Review

Today we reviewed conditional statements and booleans
We did a bunch of exercises and challenges (functions)
Learned Lists and Functions
Brainstormed for the final project

